Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when assumptions apply.
3. Explain energy and model limits clearly.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

In the same straight-line deflection model, keep sideways Δv fixed but make lead time 3 times as long. Compare shift and name a limitation.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. Label before/after diagrams and directions.

Worked reasoning: A 2 kg cart moving at 3 m/s stops in 0.5 s. Momentum change is -6 kg·m/s; average force is -12 N. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

Marzano check - circle: ① ② ③

03 Look and explain

A 1 kg cart at 4 m/s right sticks to a 1 kg cart at rest. Draw before/after and explain which quantity is conserved.

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

04 Build the solution

A 2 kg cart slows from 6 m/s right to rest in 5 s. Find momentum change size and average stopping force size.

Marzano check - circle: ① ② ③

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Continue down each column.

05 Use the idea

A report uses equal total momentum before/after to claim a collision is elastic. What additional calculation is required?

Marzano check - circle: 1 2 3

06 Choose a method

A 4 kg cart travels left at 6 m/s. Find momentum and draw its direction.

Marzano check - circle: 1 2 3

07 Exit check

A fixed impulse gives a target .04 m/s sideways velocity change. Find shift after 500 s and state a model limitation.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Return to the worked example in problem 2. Design two stopping processes for this cart that give the same impulse but different average forces. Explain the time-force tradeoff.

Marzano check - circle: 1 2 3